0066. Plus One

Easy · Math & Geometry
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Problem

- A number is stored as an array of digits, most significant first. Add one.

Intuition

- Walk from the last digit: a digit below 9 just increments and we are done; a 9 becomes 0 and the carry moves left.
- If every digit was 9, the result is 1 followed by zeros.

Brute Force

- Convert to a number: overflows for long arrays.

Optimal Approach

- For i from the end: if d[i] < 9, d[i]++ and return.
- Else d[i] = 0.
- Return a new array of length n+1 with a leading 1.

Worked Example

```
[1,2,9].

i=2: 9 -> 0, carry
i=1: 2 -> 3, return [1,3,0]
```

Java Solution

```java
public class PlusOne {
    public int[] plusOne(int[] digits) {
        for (int i = digits.length - 1; i >= 0; i--) {
            if (digits[i] < 9) { digits[i]++; return digits; }
            digits[i] = 0;
        }
        int[] res = new int[digits.length + 1];
        res[0] = 1;
        return res;
    }
}
```

Complexity

- Time: O(n). Space: O(1), or O(n) for the all-nines case.

Pitfalls

- New int[n+1] is zero-filled, so only the first slot needs setting.